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Tumor and Dosimetric Predictors of Long-Term Visual Outcomes After Proton Beam Therapy of Uveal Melanoma

A.L. Polishchuk,¹ D.H. Char,² V. Weinberg,¹ I.K. Daftari,¹ T.B. Cole,² J.M. Quivey,¹ T.L. Phillips,¹ and K.K. Mishra¹; ¹UCSF Radiation Oncology, San Francisco, CA, ²Tumori Foundation, San Francisco, CA

Purpose/Objective(s): To discern contributing tumor and critical structure dose-volume parameters affecting long-term visual outcomes after proton beam radiation therapy (PBRT) for uveal melanoma (UM) patients.

Materials/Methods: Using our prospectively maintained database, PBRT patients treated for UM by a single surgeon with visual acuity (VA) follow-up were reviewed (1996-2010). 512 patients with pre-treatment best corrected VA (BCVA) in the treated eye of 20/200 or better were evaluated for this study and patients were stratified into three groups as follows: 20/15-20/40 (n = 363), 20/50-20/100 (n = 114), and 20/200 (n = 35). Patients received 56 GyE in 4 fractions. Median VA follow-up was 6.0 years (range 0.3 - 18). Overall, at last follow-up 271 patients had vision 20/200 or better, of which 160 patients had vision 20/40 or better. Loss in VA was defined as migration into a lower VA stratum including worse than 20/200 at most-recent post-treatment assessment compared with pre-treatment; 330 patients experienced migration into a poorer VA stratum. Univariate and multivariate Cox's stepwise regression models adjusted for pretreatment VA were applied to identify independent predictors for loss in VA among patient, tumor, and dose-volume histogram (DVH) parameters.

Results: Univariate analysis indicated that loss in VA was significantly related to the following patient/tumor parameters: older age (p = 0.0009), diabetes (p = 0.009), greater tumor height (p < 0.0001), larger tumor diameter (p < 0.0001), closer tumor distance to fovea (p = 0.006) and to optic disc (p = 0.046). DVH analysis revealed that at ≥50% dose (≥28 GyE) to the lens, optic disc and macula, significant predictors of VA loss included > 30% of the lens volume, 100% of the disc and > 0% of the macula receiving this dose (p = 0.0002, 0.0001 and 0.0005, respectively). After adjusting for pretreatment VA, multivariate analysis identified the following independent predictors of loss in VA (likelihood ratio test; hazard ratio (HR), 95% confidence interval): increased tumor height (p < 0.0001; HR = 1.13 [1.08-1.18]), macula receiving ≥50% dose (p < 0.0001; HR = 1.87 [1.44-2.42] for 0 vs > 0%), increased age at the time of PBRT (p < 0.0001; HR = 1.02 [1.01-1.03]), increased tumor diameter (p = 0.02; HR = 1.06 [1.02-1.10]) and 100% of the optic disc receiving ≥50% dose (p = 0.02; HR = 1.36 [1.05-1.77]).

Conclusions: For UM patients undergoing PBRT, the change in visual acuity from pretreatment BCVA may be predicted by patient age at treatment, tumor height, largest tumor diameter, and optic disc and macula dose-volume parameters. Further analysis is planned to evaluate VA change at various time points of follow-up as well as relationships for disc and macula sparing and dose effects.

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Late Effects of Radiation Therapy for Optic Nerve Sheath Meningioma (ONSM)

L. Stalpers, Y. van Praag, and P. Saeed; Academic Medical Center - University of Amsterdam, Amsterdam, Netherlands

Purpose/Objective(s): Optic nerve sheath meningioma (ONSM) is a rare, slowly growing tumor of the anterior visual pathway. We analyzed the long-term effects of primary radiation therapy in ONSM, with special

emphasis on ophthalmological, pituitary and radiological changes, and on the quality of life (QOL).

Materials/Methods: A cross-sectional cohort study of fourteen patients with ONSM who had radiation therapy between January 2001 and February 2011. All patients had an extended ophthalmological, endocrine and radiological reassessment. The Graves' Ophthalmopathy-Quality of Life questionnaire was used to describe changes in health related quality of life. Medical ethical approval was granted (NL39633.018.12).

Results: Twelve women and two men with a median age of 58 years (range 30-65 years) at diagnosis were included in the study. After a mean and median follow-up of 74.4 and 50.5 months (range 21-141 months), visual acuity improved in 4 out of 14 (29%) patients, remained stable but functional in 3/14 (21%), and remained poor or deteriorated in 7/14 (50%). MRI tumor control rate was 84%; all eyes were preserved, and no patient required salvage treatment. Late effects included radiation-induced retinopathy (21%), dry eye (7%) and cataract (7%). None of the patients had a deterioration of pituitary function. The largest impact on QOL was due to a moderate to severe change in appearance in half of the patients, and 35-43% were limited in doing leisure activities, driving and reading.

Conclusions: The prognosis for patients treated with radiation therapy for ONSM is good, and if treated in time, visual functioning can be preserved. The negative impact on QOL in patients with ONSM is mostly due to changes in appearance by the tumor and due to limitations in daily activities from loss in vision, and only to a minor degree by long-term side-effects of radiation therapy.

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MRI-Based Verification of Plaque Positioning and Treatment Planning for Episcleral Brachytherapy

S.G. Price,¹ J. Garcia-Ramirez,² C. Bertelsman,² Y. Hu,¹ M.J. Rivard,³ K. Tanderup,¹ P. Rao,¹ P. Grigsby,¹ and J. Esthappan¹; ¹Washington University School of Medicine, St. Louis, MO, ²Barnes Jewish Hospital, St. Louis, MO, ³Tufts New England Medical Center, Boston, MA

Purpose/Objective(s): Volumetric imaging to verify plaque positioning is typically not performed for uveal melanoma patients receiving episcleral brachytherapy (EBT). Thus, clinical outcomes and toxicities have been historically related to doses based on estimates of the pre-implant geometry. A magnetic resonance imaging (MRI)-based technique was used to perform pre- and post-implant imaging and treatment planning for comparison of estimated and administered implant geometry and dose distributions.

Materials/Methods: Four patients were prescribed 85 Gy over 96 hours from COMS-based EBT with ¹²⁵I seeds. MRI (1.5 T) was performed 1 week prior to and about 3 days after implantation. T2- and T1-weighted sequences were acquired for soft tissue definition and, in post-implant scans, proton-density weighted sequences were added for improved plaque visualization. Pre- and post-implant scans were used to generate "pre-plans" and "post-plans," respectively, and fused to account for motion of the eye. In the pre-plans, a digital plaque model was positioned relative to the tumor, sclera, and nerve. In the post-plans, the same plaque model was positioned based on the signal void of the plaque. Tumor and normal tissues were contoured, and dose points were defined on the pre-implant scans. All contours and points were registered on to the post scans and adjusted if necessary. Isodoses were computed using the TG-43 formalism with no heterogeneity corrections. The shift between plaques in the fused pre- and post-plans was determined. The tilt of the plaque away from the outer sclera was measured on the post-implant images. Point doses, percentage of tumor volume receiving 85 Gy (V₁₀₀), and dose to 100% of tumor volume (D₁₀₀) were compared between pre- and post-plans.

Results: Shifts and tilts of the plaques ranged from 1.6 mm to 3.9 mm (mean 2.4 mm). Mean differences in point doses for the tumor apex, prescription depth, retina, optic disc, fovea, and lens indicated decreases from the pre- to post-implant plans of 15.1±10.8%, 17.9±10.5%, 16.6±13.5%, 28.1±12.8%, 31.5±18.4%, and 6.1±9.6%, respectively. The ranges in D₁₀₀ for pre- and post-plans were 85.8-128.3 Gy and 45.1-88.0 Gy, respectively. For three patients, the prescription depth was at least 1 mm beyond the tumor apex, there